

NANNING, China

WMO: 594310

Lat: 22.7840N

Lon: 108.5472E

Elev: 153

StdP: 99.50

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	6.1	-2.1	3.2	12.7	0.1	3.9	11.2	6.9	18.7	6.1	16.5	1.7	340	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	35.1	26.4	34.1	26.3	33.2	26.0	27.9	32.3	27.4	31.6	27.0	31.0	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.9	30.1	26.2	22.1	29.4	26.1	21.8	29.2	90.7	32.6	88.6	31.7	86.7	31.4	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.8	DB	1.9	37.2	1.5	0.8	0.9	37.8	0.0	38.3	-0.8	38.7	-1.9	39.3	(4)
			WB	1.4	28.8	1.4	0.5	0.4	29.2	-0.4	29.5	-1.2	29.7	-2.3	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	13.3	15.2	18.2	23.2	26.5	28.3	28.7	27.2	24.1	19.7	14.9		
		DBStd	6.44	4.27	4.89	4.27	3.71	2.48	1.69	1.56	1.54	2.09	2.72	3.44	3.60	(7)
		HDD10.0	24	14	7	1	0	0	0	0	0	0	0	3		
		HDD18.3	489	167	112	58	7	0	0	0	0	1	25	119		
		CDD10.0	4540	115	153	255	397	511	549	582	578	516	438	291	155	(10)
		CDD18.3	1964	9	25	54	155	252	299	323	320	266	181	66	12	(11)
		CDH23.3	18736	24	115	259	1120	2354	3257	3637	3529	2635	1374	385	46	(12)
(13)	Wind	CDH26.7	7057	1	26	61	353	872	1264	1479	1451	1032	443	72	3	(13)
		WSAvg	2.0	2.0	2.1	2.1	2.2	2.2	2.0	2.1	1.8	1.9	1.8	1.8	1.9	(14)
		PrecAvg	1658	47	39	66	90	180	310	348	252	153	77	61	36	(15)
		PrecMax	2377	172	155	169	265	376	536	662	480	335	232	223	122	(16)
		PrecMin	1265	4	4	8	18	83	73	139	80	23	5	3	3	(17)
		PrecStd	265	42	32	39	58	63	110	138	90	73	64	57	32	(18)
		DB	25.8	30.0	31.0	33.9	35.1	35.2	36.1	36.0	35.0	33.1	30.2	26.9		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	19.9	21.6	23.5	25.7	26.8	26.9	26.6	26.4	25.6	23.4	23.0	21.0		
		DB	23.8	26.5	28.2	32.0	33.5	34.2	34.9	34.9	33.9	31.8	28.7	24.5		
		MCWB	18.6	20.8	22.8	24.9	26.1	26.8	26.6	26.3	25.4	23.3	21.6	19.2		
		DB	21.9	24.2	26.1	30.2	32.2	33.1	33.9	33.9	32.9	30.6	27.2	22.9		
		MCWB	17.8	19.4	21.7	24.2	25.3	26.5	26.5	26.3	25.1	22.9	20.5	17.7		
		DB	20.0	22.1	24.1	28.8	31.1	32.0	32.8	32.8	31.8	29.2	25.8	21.1		
		MCWB	16.8	18.9	20.8	23.5	25.1	26.2	26.5	26.3	24.8	22.2	19.9	16.8		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	20.9	22.4	24.3	26.5	28.0	28.5	28.5	28.2	27.2	25.8	24.5	22.7		
		MCDB	24.3	27.8	29.2	31.8	33.3	33.4	33.4	32.7	31.3	29.8	27.7	25.6		
		WB	19.7	21.4	23.2	25.6	27.1	27.7	27.7	27.6	26.7	25.1	23.6	21.0		
		MCDB	22.3	25.4	27.3	30.4	31.7	31.9	32.1	31.8	30.7	28.9	26.6	23.3		
		WB	18.7	20.4	22.4	24.9	26.5	27.2	27.2	27.1	26.3	24.5	22.7	19.5		
		MCDB	20.9	23.4	25.5	29.1	30.5	31.1	31.2	31.2	30.1	28.1	25.5	21.5		
		WB	17.5	19.4	21.4	24.2	26.0	26.8	26.8	26.7	25.9	23.9	21.7	17.6		
(34)	Mean Daily Temperature Range	MCDB	19.6	21.7	23.8	27.7	29.5	30.3	30.5	30.5	29.6	27.3	24.3	20.3		
		MDBR	6.9	6.9	6.5	7.4	7.7	7.1	7.2	7.7	8.2	8.7	9.0	8.5		
		5% DB	10.2	10.6	10.1	10.0	9.7	8.9	9.4	9.5	9.9	10.5	11.0	11.3		
		MCWBR	5.5	5.3	5.1	4.3	3.7	3.1	2.9	3.0	3.1	3.6	4.8	5.8		
		5% WB	8.3	9.7	9.2	9.1	8.4	7.6	8.0	8.3	8.5	8.4	8.3	8.4		
		MCWBR	5.0	5.4	5.3	4.3	3.7	3.1	2.9	3.1	3.1	3.4	4.5	5.4		
		taub	0.616	0.644	0.792	0.751	0.575	0.549	0.522	0.591	0.635	0.721	0.608	0.627		
(40)	Clear-Sky Solar Irradiance	taud	1.675	1.655	1.425	1.541	1.956	2.046	2.115	1.904	1.769	1.561	1.744	1.674		
		Ebn at Noon	644	664	592	632	750	764	785	734	691	598	647	614		
		Edh at Noon	224	242	318	287	189	171	160	197	222	262	207	217		
		RadAvg	2.13	2.42	2.65	3.48	4.19	4.22	4.63	4.73	4.48	3.89	3.10	2.49		
(44)	All-Sky Solar Radiation	RadStd	0.53	0.45	0.39	0.42	0.39	0.55	0.45	0.28	0.29	0.32	0.49	0.47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.28	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A	+62

Nomenclature: See separate page